

Adaptive Heuristic Portfolio TSP Report

Overview

- Condition count: 8.
- Best held-out TSPLIB gap: phase8_best_single_fixed_heuristic.
- Best combined transfer gap: phase8_oracle_selector.
- Lowest selector regret: phase8_best_single_fixed_heuristic.
- Pareto-efficient conditions: phase8_full_solver_evolution, phase8_llm_evolved_adaptive_controller, phase8_llm_evolved_controller_diversity_failure_replay, phase8_supervised_ml_selector.

Run Metadata

- run_name: run_20260520_172645_c
- started_at_local: 2026-05-20 17:26:45
- finished_at_local: 2026-05-20 17:55:40
- duration_hhmm: 00:29
- duration_seconds: 1735.193
- seed_offset: 2000
- replicate_label: c
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Final TSPLIB Gap	Selector Regret	Runtime (ms)	Runtime-Adjusted Gap	Family Gap	Transfer Gap	Mean Novelty	Mean Complexity	Pareto Efficient
phase8_best_single_fixed_heuristic	best_fixed	none	0.07226	0.0	1491.554171	0.07226	0.007284	0.031222	0.0	0.0	False
phase8_random_portfolio	random_portfolio	none	0.190005	0.117745	1482.355357	0.190005	0.082535	0.122129	0.0	0.0	False
phase8_oracle_selector	oracle_selector	none	0.07226	0.0	1535.363729	0.074382	8.9e-05	0.026678	0.0	0.0	False

Condition	Mode	Replay	Final TSPLI B Gap	Selector Regret	Runtime (ms)	Runtime -Adjusted Gap	Family Gap	Transfer Gap	Mean Novelty	Mean Complexity	Pareto Efficient
phase8_supervised_ml_selector	supervised_selector	none	0.07226	0.0	1465.688143	0.07226	0.007284	0.03122	0.0	0.0	True
phase8_llm_static_selector	static_selector	none	0.214311	0.142051	488.999614	0.214311	0.024801	0.094621	0.0	0.62	False
phase8_llm_evolved_adaptive_controller	adaptive_controller	none	0.203245	0.130985	344.320514	0.203245	0.027485	0.092239	0.044633	1.7025	True
phase8_llm_evolved_controller_diversity_failure_replay	adaptive_controller	diversity_failure	0.204775	0.132515	267.676329	0.204775	0.059962	0.113314	0.0	0.62	True
phase8_full_solver_evolution	full_solver	none	0.213387	0.141127	206.0418	0.213387	0.150269	0.173523	0.814332	0.56	True

Condition Notes

phase8_best_single_fixed_heuristic

- Execution mode best_fixed with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.07226.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 1491.554171 ms, runtime inflation 0.0, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.
- Fixed heuristic choice: farthest_insertion_two_opt.

phase8_random_portfolio

- Execution mode random_portfolio with replay none.
- Final held-out gap 0.190005, selector regret 0.117745, runtime-adjusted gap 0.190005.
- Family transfer 0.082535 and combined transfer 0.122129.
- Runtime 1482.355357 ms, runtime inflation -0.006167, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.

phase8_oracle_selector

- Execution mode oracle_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.074382.

- Family transfer 8.9e-05 and combined transfer 0.026678.
- Runtime 1535.363729 ms, runtime inflation 0.029372, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.

phase8_supervised_ml_selector

- Execution mode supervised_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.07226.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 1465.688143 ms, runtime inflation -0.017342, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: True.

phase8_llm_static_selector

- Execution mode static_selector with replay none.
- Final held-out gap 0.214311, selector regret 0.142051, runtime-adjusted gap 0.214311.
- Family transfer 0.024801 and combined transfer 0.094621.
- Runtime 488.999614 ms, runtime inflation -0.672154, mean novelty 0.0, and complexity 0.62.
- Pareto efficient: False.
- Controller signature: nearest_neighbor_multistart:farthest_insertion_two_opt:farthest_insertion_two_opt:edge_preserving_restart:stag3:cand3:restart1.
- Controller rule summary: {"acceptance_bias": 0.006, "candidate_limit_offset": 3, "clustered_heuristic": "farthest_insertion_two_opt", "corridor_heuristic": "limited_three_opt", "default_heuristic": "farthest_insertion_two_opt", "description": "Static instance-conditioned portfolio selector using interpretable structure cues; schedules small bounded perturbation parameters via fixed offsets and thresholds.", "failed_perturbation_threshold": 2, "grid_like_heuristic": "farthest_insertion_two_opt", "low_improvement_threshold": 0.08, "name": "deterministic_instance_adaptive_frozen_portfolio_v1", "nearest_neighbor_trap_heuristic": "annealed_multi_start", "random_like_heuristic": "nearest_neighbor_multistart", "restart_offset": 1, "stagnation_switch_heuristic": "edge_preserving_restart", "stagnation_threshold": 3, "temperature_scale": 1.05, "time_budget_trigger": 0.55, "two_cluster_bottleneck_heuristic": "farthest_insertion_two_opt"}

phase8_llm_evolved_adaptive_controller

- Execution mode adaptive_controller with replay none.
- Final held-out gap 0.203245, selector regret 0.130985, runtime-adjusted gap 0.203245.
- Family transfer 0.027485 and combined transfer 0.092239.
- Runtime 344.320514 ms, runtime inflation -0.769153, mean novelty 0.044633, and complexity 1.7025.
- Pareto efficient: True.
- Controller signature: nearest_neighbor_multistart:cluster_first_local_search:farthest_insertion_two_opt:annealed_multi_start:stag4:cand-1:restart2.
- Controller rule summary: {"acceptance_bias": 0.006, "candidate_limit_offset": -1, "clustered_heuristic": "cluster_first_local_search", "corridor_heuristic": "edge_preserving_restart", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-adaptive hyper-heuristic controller selecting from a frozen portfolio using interpretable structural descriptors and online stagnation/counters to schedule bounded adjustments (candidate list limit, restarts, and annealing temperature/ac", "determinism": {"no_random_choice": true, "rule_evaluation_order": "first_match_wins", "seed_free": true}, "failed_perturbation_threshold": 3,

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"grid_like_heuristic": "candidate_pruned_two_opt", "low_improvement_threshold": 0.08, "name":
"tsp_frozen_portfolio_instance_adaptive_v2", "nearest_neighbor_trap_heuristic": "candidate_pruned_two_opt",
"policy": {"fallback": {"tuning": {"candidate_limit_offset": 0, "restart_offset": 0}, "use": "default_heuristic"},
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low_improvement_threshold"}, {"tuning": {"restart_offset": "restart_offset"}, "use": "random_like_heuristic", "when":
"failed_perturbation_count >= failed_perturbation_threshold and time_budget_used < time_budget_trigger"},
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stagnation_threshold and time_budget_used >= time_budget_trigger"}], "static_structure_rules": [{"use":
"grid_like_heuristic", "when": "grid_likeness >= 0.45 and dimension >= 80"}, {"use":
"two_cluster_bottleneck_heuristic", "when": "bottleneck_score >= 0.52 or two_cluster_bottleneck_score >= 0.52"},
{"use": "clustered_heuristic", "when": "clustering_score >= 0.46 and cluster_separation >= 0.46"}, {"use":
"corridor_heuristic", "when": "corridor_score >= 0.47"}, {"use": "nearest_neighbor_trap_heuristic", "when":
"nearest_neighbor_trap_score >= 0.52 and nn_distance_cv >= 0.55"}], "random_like_heuristic":
"nearest_neighbor_multistart", "restart_offset": 2, "stagnation_switch_heuristic": "annealed_multi_start",
"stagnation_threshold": 4, "temperature_scale": 1.25, "time_budget_trigger": 0.7, "two_cluster_bottleneck_heuristic":
"farthest_insertion_two_opt"}

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phase8_ilm_evolved_controller_diversity_failure_replay

- Execution mode adaptive_controller with replay diversity_failure.
- Final held-out gap 0.204775, selector regret 0.132515, runtime-adjusted gap 0.204775.
- Family transfer 0.059962 and combined transfer 0.113314.
- Runtime 267.676329 ms, runtime inflation -0.820539, mean novelty 0.0, and complexity 0.62.
- Pareto efficient: True.
- Controller signature: nearest_neighbor_multistart:cluster_first_local_search:edge_preserving_restart:limited_three_opt:stag3:cand3:restart1.
- Controller rule summary: {"acceptance_bias": 0.0, "candidate_limit_offset": 3, "clustered_heuristic": "cluster_first_local_search", "corridor_heuristic": "candidate_pruned_two_opt", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-adaptive selection among a frozen TSP portfolio; bottleneck and clustered layouts trigger specialized restart or clustering; stagnation triggers a limited 3-opt escalation; budget trigger shifts toward faster candidate-pruned local search", "failed_perturbation_threshold": 2, "grid_like_heuristic": "cheapest_insertion_two_opt", "low_improvement_threshold": 0.08, "name": "interpretable_instance_adaptive_heuristic_switchboard_v1", "nearest_neighbor_trap_heuristic": "candidate_pruned_two_opt", "random_like_heuristic": "nearest_neighbor_multistart", "restart_offset": 1, "stagnation_switch_heuristic": "limited_three_opt", "stagnation_threshold": 3, "temperature_scale": 1.15, "time_budget_trigger": 0.65, "two_cluster_bottleneck_heuristic": "edge_preserving_restart"}

phase8_full_solver_evolution

- Execution mode full_solver with replay none.
- Final held-out gap 0.213387, selector regret 0.141127, runtime-adjusted gap 0.213387.
- Family transfer 0.150269 and combined transfer 0.173523.
- Runtime 206.0418 ms, runtime inflation -0.861861, mean novelty 0.814332, and complexity 0.56.
- Pareto efficient: True.

Judge Appendix

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Conservative Interpretation of TSP Adaptive Heuristic Portfolio Results

Primary Endpoint: Held-out TSPLIB Gap

- The best held-out TSPLIB gaps are reached by:
- **phase8_best_single_fixed_heuristic** and **phase8_supervised_ml_selector**, both achieving a gap of **0.07226**.
- The **oracle selector** matches this gap (0.07226) and serves as an upper bound.
- Other adaptive controllers (**phase8_llm_evolved_adaptive_controller**, **phase8_llm_evolved_controller_diversity_failure_replay**) have higher TSPLIB gaps (~0.20), indicating less optimal held-out performance despite adaptation.
- The random portfolio and static selector perform worse (TSPLIB gaps 0.19 and 0.21 respectively).

Selector Regret vs Oracle Portfolio

- The oracle selector by definition has **0 regret**.
- The **best_fixed_heuristic** and **supervised_ml_selector** also report zero selector regret, implying no performance loss relative to oracle on selector choice.
- Adaptive controllers have selector regrets around 0.13, indicating moderate inefficiency relative to oracle selection, despite adaptation.
- The random portfolio and static selector show highest regret (>0.13 and 0.14 respectively).

Runtime-Adjusted Gap & Runtime Inflation

- Runtime-adjusted gap balances solution quality and runtime:
- Best single fixed heuristic and supervised ML selector have ~0.072 runtime-adjusted gap with near-zero or slightly negative runtime inflation (no runtime increase).
- Oracle selector has slightly higher runtime-adjusted gap (0.074) with 3% runtime inflation.
- Adaptive controllers have higher runtime-adjusted gaps (~0.20) but **much lower runtime inflation** (around -0.7 to -0.8), indicating faster runtimes but at quality cost.
- Static selector also faster (-0.67 inflation) but worse quality.
- This tradeoff suggests adaptive controllers yield faster but lower-quality solutions.

Pareto Efficiency

- Pareto-efficient conditions include:
- **phase8_supervised_ml_selector** (good TSPLIB gap, low regret, no runtime inflation)
- **phase8_llm_evolved_adaptive_controller**, **phase8_llm_evolved_controller_diversity_failure_replay**, and **phase8_full_solver_evolution**
- Despite higher gaps, adaptive controllers are Pareto efficient due to better runtime efficiency or complexity.
- Best fixed heuristic and oracle selector are **not** Pareto efficient under multi-objective consideration.

Cross-Family Transfer

- Transfer gaps (performance gap when tested on different or held-out instance families):
- Best fixed heuristic & supervised ML selector: ~0.0312 (low transfer gap)

- Oracle selector: ~0.027 (lowest)
- Adaptive controllers: ~0.09-0.11 (higher but still moderate)
- Random portfolio and static selector: higher (0.10-0.12+)
- Indicates moderate generalization ability of adaptive selectors but best fixed and supervised ML selectors transfer better.

Interpretability and Instance-Adaptive Control

- Supervised ML selector is interpretable and instance-adaptive, achieves oracle-level TSPLIB gap without runtime inflation, supporting interpretable instance-adaptive control claims.
- Adaptive controllers apply interpretable, deterministic rules with bounded adjustments; however, their TSPLIB gap and selector regret are worse, albeit with runtime benefits.
- Static selector is interpretable but suffers from higher TSPLIB gap and regret.
- No evidence of discovering new heuristics; all control hyper-heuristic over known heuristics as intended.

Summary

Condition	TS PLI B Gap	Select or Re gret	Runtime- Adjusted Gap	Runtim e Inflati on	Tran sfer Gap	Pareto Efficie nt	Notes
phase8_best_single_fixed_heuristic	0.07226	0.0	0.07226	0.0	0.03122	No	Best held-out gap; baseline fixed heuristic
phase8_supervised_ml_selector	0.07226	0.0	0.07226	-0.0173	0.03122	Yes	Interpretable instance-adaptive selector; near oracle quality & runtime
phase8_oracle_selector	0.07226	0.0	0.07438	0.029	0.02668	No	Upper bound, non-deployable oracle
phase8_llm_evolved_adaptive_controller	0.20325	0.131	0.20325	-0.77	0.0922	Yes	Adaptive controller with runtime improvement but quality loss
phase8_llm_evolved_controller_diversity_failure_replay	0.20477	0.133	0.20477	-0.82	0.1133	Yes	Similar adaptive controller tradeoffs
phase8_full_solver_evolution	0.21338	0.141	0.21338	-0.86	0.1735	Yes	Highest complexity, low runtime, worst quality
phase8_llm_static_selector	0.21431	0.142	0.21431	-0.67	0.0946	No	Static interpretable selector but worse quality/runtime tradeoff
phase8_random_portfolio	0.19000	0.1177	0.19000	~0.0	0.1221	No	Baseline random selection; worst performance

Recommendations

- The ****supervised ML selector**** offers the best balance of held-out TSPLIB gap, zero selector regret, moderate runtime efficiency, and interpretable instance-adaptive control. This makes it the preferable strategy for deployment.
- Adaptive controllers achieve significant runtime savings but at cost of solution quality and increased selector regret; careful tuning needed before practical use.
- Oracle selector remains a useful theoretical upper bound but not deployable.
- No new heuristic discovery claims are warranted; results focus on hyper-heuristic control improving heuristic portfolio selection.
- Further work could focus on reducing selector regret of adaptive controllers while maintaining runtime advantages, to approach oracle performance more closely.